

**Proposed Data Centre
Development at No. 2-10,
Tai Yuen Street, Kwai
Chung, N.T.**

Technical Specification for Steel
Reinforcement

Revision: 0

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1 General

- a) Reinforcement for concrete shall be ribbed steel bar, cold reduced steel wire or steel fabric of suitable composition, manufacture, and chemical and physical properties.
- b) Ribbed steel reinforcing bars shall comply with BS 4449:2005+A2:2009 and CS2:2012.
- c) Plain steel reinforcing bars up to 12 mm diameter shall comply with BS 4482:2005.
- d) Unless specified on Drawings, hard drawn plained steel wires and cold worked steel bars shall not be used without the approval of the Architect.

2 Storage and Conditions of Reinforcement

- a) Reinforcement shall be stored in clean and dry conditions off the ground in a manner which will not result in damage, deformation or contamination to the reinforcement. Fabric reinforcement shall be stored horizontally.
- b) Reinforcement shall not be stored on or adjacent to concrete surfaces which form part of the permanent works. Different types and sizes of reinforcement shall be stored separately.
- c) Reinforcement shall be clean and free from loose mill scale, loose rust, oil or grease at the time of fixing in position and immediately before concreting.
- d) Reinforcement which in the opinion of the Architect has been seriously contaminated or heavily rusted shall be removed off site unless it has been cleaned to the satisfaction of the Architect.

3 Cutting and Bending of Reinforcement

- a) Bending schedules complying with BS4466 and Code of Practice for Structural Use of Concrete 2013 shall be prepared by the Contractor and submitted to the Architect before bending of reinforcement starts.
- b) Reinforcement shall be cut and bent to the dimensions as derived from the Drawings and in accordance with BS 4466 & Code of Practice for Structural Use of Concrete.
- c) Reinforcement shall be cut and bent at temperatures between 5°C and 100° C.
- d) Ribbed bar or cold worked steel bars shall not be rebent or straightened after having been bent. Where it is necessary to bend plained steel bars projecting from concrete, the internal radius of bend shall not be less than twice the bar diameter.
- e) Reinforcement shall not be bent or straightened in a manner that in the opinion of the Architect, will damage the concrete or the reinforcement including the connectors. Bars with kinks or bends not shown on Drawings shall not be used.

4 Fixing of Reinforcement

- a) Bar reinforcement, fabric reinforcement and reinforcement connector for tension joints from each batch shall not be fixed until testing of the batch has been completed and found satisfactory.
- b) Reinforcement shall be fixed into position and secured against displacement. It shall be secured adequately with 1.6 mm diameter annealed iron tying wires or other approved tying devices and clips. The ends of the tying wires, tying devices or clips shall be turned into the body of concrete and not encroach into the concrete cover.
- c) Adequate concrete cover spacers, steel chairs, or other approved cover spacers shall be placed at a maximum spacing of 1.5 m to maintain the specified concrete cover at all positions.
- d) Concrete cover blocks shall be made as small as practical and be of such shape that they will not overturn when concrete is placed. They shall be made of concrete with 10 mm maximum aggregate

size and having the same strength as the adjacent concrete. Tying wires shall be cast in the cover blocks for tying them to the reinforcement.

- e) Steel chairs shall be provided with concrete cover blocks when they are resting on soffit formworks.
- f) Other spacers if used shall be of such materials and design that will not lead to corrosion of reinforcement or spalling of the concrete cover.
- g) The concrete cover to reinforcement specified on Drawings shall be the nominal cover to all reinforcement bars and shall be adhered to during construction with a permissible deviation of +5 mm and 0 mm.
- h) Prefabricated reinforcement cages shall be adequately supported and braced before lifting.
- i) Reinforcement which is free-standing shall be secured in position and braced to prevent movement due to wind and other loads.

5 Laps and Splices of Reinforcement

- a) Laps and splices shall only be made by the methods specified and at the positions shown on the Drawings, or with the prior approval of the Architect.
- b) Reinforcement connectors shall be fixed in accordance with the manufacturer's recommendations and using equipment recommended by the manufacturer.

6 Welding of Reinforcement

- a) When welding of reinforcement is specified on the Drawings, the Contractor shall submit the welding procedures for the approval of the Architect before such welding is carried out.
- b) No other welding of reinforcement shall be allowed.

7 Mechanical Splicing of Reinforcement

- a) Where the use of mechanical splicing is specified on the Drawings or is proposed by the Contractor and approved by the Architect, its use shall comply with the following subclauses. Particulars of the proposed material and method of installation for the reinforcement connectors, including the manufacturer literature, shall be submitted to the Architect at least 28 days before fixing of the reinforcement connectors starts. The Architect may require trial tests of each type of proposed connectors prior to commencement of works, the trial test shall use the same method of installation and material as the working connectors.
- b) Splices shall be staggered so that no more than half of the bars are spliced at any cross section of the structural member.
- c) The splicing shall be carried out in strict compliance with the supplier's specification. It shall be carried out by persons fully trained by the supplier in the correct operation and procedures and using equipment and tools as specified by the supplier.
- d) The spacing of the bars to be spliced shall be checked by the supplier to ensure that it is sufficient and suitable for the splicing to be properly carried out.
- e) At least one sample for every 100 splices shall be tested. The Architect may order additional tests to be carried out to ensure the quality and workmanship at all times.
- f) Test results shall be submitted to the Architect within 7 days after testing and prior to commencement of pouring concrete to the area of works of which samples are being tested.

8 Inspection of Reinforcement

- a) No concrete shall be placed to any part of the reinforced concrete work until the reinforcement has been inspected and approved by the Architect.
- b) Ample notice of 24 hours in advance, or such shorter period as agreed by the Architect, shall be given to the Architect for inspection of the completed fixing of reinforcement.

9 Protection of Reinforcement

- a) Where starter bars or projecting reinforcement are to be left exposed for a prolonged period, they shall be protected by coating thoroughly but thinly with cement slurry, dense tape, or other approved methods.
- b) Reinforcement shall not be contaminated or displaced as a result of access over the reinforcement; access shall be obtained by using planks and ladders or by other method agreed by the Architect.

10 Testing of Ribbed Steel Bars

- a) Every batch of Ribbed steel bars delivered to Site shall be covered by:
 - i. certificate of origin from the local supplier, indicating the source of steel and the quantity to be delivered;
 - ii. manufacturer's mill certificate to certify its compliance to BS 4449 and CS2:2012; and
 - iii. report of tests carried out by an accredited HOKLAS laboratory.
- b) The above three items of (1)(a), (b) & (c) shall be submitted in one lot for each batch within 21 days after the steel bars are delivered to site and at least 14 days before fixing. The submission of these documents shall be accompanied by a record of reinforcement delivery in the specified format as annexed with this Specification.
- c) A batch is defined as any quantity of bars of one size and grade, from one mill, covered by one mill certificate, and delivered to site at any one time. Steel bars in more than one delivery within a short period may be considered as part of the same batch if the deliveries belong to one single consignment from the same manufacturer and are covered by the same mill certificate.
- d) QA Stockists are now required to be certified to the requirements of ISO 9001 by a certification body accredited by Hong Kong Accreditation Service with a scope related to good practices on storage, handling and supply of steel reinforcing bars in accordance with the specific requirements stipulated in this Construction Standard.

In addition, QA Stockists are also required to implement specific control on non-conforming products as stated in Cl. 4.2 of this CS2: 2012.
- e) A stockist is an organisation that receives steel reinforcing bars from a manufacturer or another stockist, and performs the requirements in accordance with this Standard. All stockists shall be Quality Assured (QA) Stockists. A QA Stockist is a stockist certified to the requirements of ISO 9001 by a certification body accredited by HKAS for its quality management system with a scope related to good practices of storage, handling and supply of steel reinforcing bars to a purchaser in accordance with this Standard.
- f) A series of tests is required to be carried out on all reinforcement delivered to site before used in the construction work. The frequency of testing shall depend on the classification of the reinforcement. Section 5 of CS2 shall be referred to for details of testing.
- g) The specimens shall be selected from different bars in the batch. The number of samples to be provided from each batch shall be as stated in Table 1.

1. **Table 1 – Rate of Test and Number of Test Specimens per Batch**

Description	No. of test specimens per batch											
	Class 1						Class 2					
	Size of batch	Tensile	Rebend	Mass per metre	Chemical composition (product analysis)	Bond property	Size of batch	Tensile	Rebend	Mass per metre	Chemical composition (product analysis)	Bond property
Steel reinforcing bar nominal size 6 mm - 16 mm	0 - 60 tonnes	3	1	3	1	1	0 - 35 tonnes	3	1	3	1	1
	each additional 60 tonnes or part of 60 tonnes	1	Nil	1	Nil	Nil	each additional 35 tonnes or part of 35 tonnes	3	Nil	3	Nil	Nil
Steel reinforcing bar nominal size 20 mm - 32 mm	0 - 80 tonnes	3	1	3	1	1	0 - 45 tonnes	3	1	3	1	1
	each additional 80 tonnes or part of 80 tonnes	1	Nil	1	Nil	Nil	each additional 45 tonnes or part of 45 tonnes	3	Nil	3	Nil	Nil
Steel reinforcing bar nominal size exceeding 32 mm	0 - 100 tonnes	3	1	3	1	1	0 - 55 tonnes	3	1	3	1	1
	each additional 100 tonnes or part of 100 tonnes	1	Nil	1	Nil	Nil	each additional 55 tonnes or part of 55 tonnes	3	Nil	3	Nil	Nil

- h) All verification tests imposed under Buildings Ordinance section 17(1)6 shall be the tests referred to in CS2:2012 and shall be performed by a Hong Kong Laboratory Accreditation Scheme (HOKLAS) accredited laboratory.
- i) The test report shall include the following information:
- Classification and Certification by QA Stockists for Class 1, Class 2 and Class 3 reinforcement;
 - Full description of test specimens including name of steel manufacturer, country of origin, grade and size as stated by the person submitting them for testing;
 - Sketch of pattern of deformation and mill marks;
 - Identification of the batch and date of delivery to the site to which the test specimens relate;
 - Measured mass and calculated effective cross sectional area of the test specimens; and
 - Results of yield stress, tensile strength, elongation, bend and rebend tests if required.
- j) Corresponding certificate shall be submitted with the test reports. For Class 1 and Class 2 reinforcement, the QA Stockist shall issue the certificate complying with CS2:2012. For Class 3 reinforcement, the Supplier shall issue the certificate complying with CS2:2012.
- k) The test results shall comply with the requirements of BS 4449:1988 and CS2:2012, and in particular, the tensile properties specified in Table 2.

Table 2 – Tensile Property Requirement

Grade	Yield strength, R_e (MPa)	Tensile/yield strength ratio, R_m/R_e	Total elongation at maximum force, A_{gt} (%)
250	250	1.15	5.0
500B	500	1.08	5.0
500C	500	≥ 1.15 and < 1.35	7.5
NOTES: 1. Values of R_e specified are characteristic with $p = 0.95$. 2. Values of R_m/R_e specified are characteristic with $p = 0.90$. For grade 500C steel reinforcing bar, the upper limit of R_m/R_e is 1.35. 3. Values of A_{gt} specified are characteristic with $p = 0.90$. 4. Values of R_m and R_e are calculated using the nominal cross-sectional area.			

The absolute maximum permissible value of yield strength of grade 500 steel reinforcing bar is 650 MPa.

For yield strength (R_e) of grade 500 steel reinforcing bar, the upper yield strength (R_{eH}), which is the maximum value of stress prior to the first decrease in force, shall apply. R_e shall be determined from the 0.2% proof strength ($R_{p0.2}$) in accordance with Appendix A if a yield phenomenon is not present. For Grade 250 steel reinforcing bars, R_e shall be determined from $R_{p0.2}$.

- l) Any batch if deemed not to comply shall be removed from the Site totally.

11 Testing of Steel Fabric

- Every batch of steel fabric delivered to the Site shall be tested to verify the mass, bend and tensile properties in accordance with the Tests specified in BS 4483.
- Each sample of fabric reinforcement shall also be tested to determine the weld shear strength.
- The test report shall include the tensile test results together with the mill certificates, results of routine inspection and testing provided by the manufacturer.

12 Testing of Hard Drawn Steel Wires and Cold Worked Steel Bars

- a) Every batch of hard drawn steel wire and cold worked steel bars, when specified or approved to be used, delivered to the Site shall be tested to verify the mass, bend and tensile properties in accordance with the Tests specified in BS 4482 and BS 4461 respectively.
- b) The test report shall include the tensile test results together with the mill certificates, results of routine inspection and testing provided by the manufacturer.

13 Testing of Reinforcement Connectors

- a) Testing of Reinforcement Connectors

14 Number of Test on Each Sample

- a) The number of tests on each sample other than ribbed steel bars shall be as stated in Table 3.

Table 3 - Number of Tests on Each Sample of Reinforcement

Description	Type and number of tests							
	Tensile	Bend	Rebend	Unit Mass	Weld Shear Stress	Thickness , Adhesion and Continuity	Weight and Uniformity of Galvanized Coating	Pitch, Dimension
Cold worked steel bar	3	-	12	3	-	-	-	-
Cold reduced steel wire	3	-	1	3	-	-	-	-
Steel fabric -fabric sheet -longitudinal wire -transverse wire	-	-	-	3	1	-	-	-
	3	-	1	-	-	-	-	1
	3	-	1	-	-	-	-	1
Reinforcement connectors for tension joints	3	-	-	-	-	-	-	-

15 Record of Reinforcement Delivery

Project Ref : _____

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D/O No.	Delivery Date	Bar Dia	Bar Pattern	Mill Cert No.	Test Report No.

Document prepared by

Aurecon Hong Kong Limited

Unit 1608, 16/F, Tower B,
Manulife Financial Centre,
223 – 231 Wai Yip Street,
Kwun Tong, Kowloon,
Hong Kong

T +852 3664 6888

F +852 3664 6999

E hongkong@aurecongroup.com

W aurecongroup.com

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Aurecon offices are located in:

Angola, Australia, Botswana, China,
Ghana, Hong Kong, Indonesia, Kenya,
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